

Triangles

Chapter 6 - Hinglish Notes, Theorems, Examples + Solutions

Ye Chapter 6 (Triangles) ka complete guide hai. Pehle red-page ke core basics, fir chapter ka main content - similar triangles, criteria (AA/SSS/SAS), BPT/Thales aur Pythagoras. End me solved examples aur 20 practice questions ke step-by-step solutions hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

PART A - Core Basics (Miss Mat Karna)

1. Triangle

- 3 side aur 3 angle hote hain.
- Teeno angle ka sum hamesha = 180 degree.

2. Ratio aur Proportion

EXAM * Exam me CONSTANTLY aata hai

Ratio: Do cheezon ki tulna, jaise $4/6 = 2/3$.

Proportion: Do ratio barabar hon, jaise $a/b = c/d$.

3. Congruent vs Similar (Bahut Important Fark)

EXAM * Exam me CONSTANTLY aata hai

Congruent: Bilkul same - shape AUR size dono same.

Similar: Same shape, par size alag (sides PROPORTIONAL).

- Yaad: photo aur uski zoom copy = similar (shape same, size alag).

4. Corresponding (Aamne-saamne wale)

EXAM * Exam me CONSTANTLY aata hai

- Similar triangles me corresponding angles equal hote hain.
- Aur corresponding sides proportional hote hain.
- Order important hai: $ABC \sim PQR \rightarrow A=P, B=Q, C=R$.

5. Parallel Lines (||)

- Do lines jo kabhi nahi milti. Symbol: ||.
- BPT (Thales) theorem inhi par based hai.

6. Square / Square Root

- $a^2 = a \times a$. $\sqrt{25} = 5$.
- Pythagoras theorem me squares use hote hain.

7. Proportion Solve Karna (Cross Multiply)

EXAM * Exam me CONSTANTLY aata hai

Rule: $a/b = c/d \rightarrow a \times d = b \times c$.

- Isi cross-multiply se unknown side nikalti hai.

PART B - Chapter 6 ka Main Content

8. Similar Triangles

EXAM * Exam me CONSTANTLY aata hai

Definition Do triangles similar hote hain jab unke corresponding angles equal hon AUR corresponding sides ek hi ratio (proportional) me hon.

- Symbol: $\sim$ (jaise $ABC \sim PQR$).
- Sides: $AB/PQ = BC/QR = CA/RP$.

9. Similarity ke Criteria

EXAM * Exam me CONSTANTLY aata hai

AA (ya AAA): Do angles equal ho to triangles similar hain.

SSS: Teeno corresponding sides ka ratio same ho.

SAS: Do sides proportional + unke beech ka angle equal ho.

10. Basic Proportionality Theorem (BPT / Thales)

EXAM * Exam me CONSTANTLY aata hai

Statement: Agar ek line triangle ki ek side ke parallel ho aur baaki do sides ko cut kare, to wo unhe SAME ratio me divide karti hai.

Agar $DE \parallel BC$ ho, to:

$$AD / DB = AE / EC$$

- Converse: agar $AD/DB = AE/EC$ ho, to $DE \parallel BC$.

11. Pythagoras Theorem

EXAM * Exam me CONSTANTLY aata hai

Right-angle triangle me:

$$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$$

Hypotenuse = right angle ke saamne wali sabse lambi side.

- Converse: agar $c^2 = a^2 + b^2$ ho, to triangle right-angled hai.
- Famous triplets: (3,4,5), (5,12,13), (8,15,17).

12. Areas of Similar Triangles

EXAM * Exam me CONSTANTLY aata hai

Rule: Do similar triangles ke areas ka ratio = unki corresponding sides ke ratio ka SQUARE.

- $\text{Area}_1 / \text{Area}_2 = (\text{side}_1 / \text{side}_2)^2$.
- Example: sides ratio 2:3 $\rightarrow$ areas ratio 4:9.

Quick Revision

Similar: angles equal + sides proportional (-).

Criteria: AA, SSS, SAS.

BPT: $DE \parallel BC \rightarrow AD/DB = AE/EC$.

Pythagoras: $\text{hyp}^2 = \text{base}^2 + \text{perp}^2$.

Areas ratio = (sides ratio)².

Solved Examples - Samjho Kaise Solve Karte Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain.

Example 1 (Pythagoras): base=3, perpendicular=4, hypotenuse?

- Step: $\text{hyp}^2 = \text{base}^2 + \text{perp}^2 = 3^2 + 4^2$.
- Step: $= 9 + 16 = 25$. $\text{hyp} = \sqrt{25}$.

Answer: Hypotenuse = 5

Example 2 (Pythagoras): hypotenuse=13, base=5, perpendicular?

- Step: $13^2 = 5^2 + p^2 \rightarrow 169 = 25 + p^2$.
- Step: $p^2 = 144 \rightarrow p = \sqrt{144}$.

Answer: Perpendicular = 12

Example 3 (Converse): Kya 6, 8, 10 right triangle banata hai?

- Step: Sabse badi side = 10. Check: $6^2 + 8^2$.
- Step: $= 36 + 64 = 100 = 10^2$. Sahi!

Answer: Haan, right-angled

Example 4 (Similarity): 2 triangles me 2-2 angles equal hain.

- Step: Do angles equal $\rightarrow$ AA criterion lagta hai.

Answer: Triangles similar hain (AA)

Example 5 (BPT): $DE \parallel BC$, $AD=2$, $DB=3$, $AE=4 \rightarrow EC$?

- Step: BPT: $AD/DB = AE/EC$.
- Step: $2/3 = 4/EC \rightarrow EC = (4 \times 3)/2$.

Answer: $EC = 6$

Example 6 (Similar sides): Similar triangles, sides 4 & 6 correspond. 5 ke corresponding?

- Step: Ratio = $4/6 = 2/3$.
- Step: 5 ko ratio se: $5 \times (3/2) = 7.5$.

Answer: 7.5

Example 7 (Area ratio): Similar triangles, sides ratio 2:3. Areas ratio?

- Step: Areas ratio = (sides ratio)².
- Step: $= 2^2 : 3^2 = 4 : 9$.

Answer: 4 : 9

Example 8 (Cross multiply): $a/4 = 6/8$, a?

- Step: Cross multiply: $a \times 8 = 4 \times 6$.
- Step: $8a = 24 \rightarrow a = 24/8$.

Answer: $a = 3$

Example 9 (Similar sides): $ABC \sim PQR$, sides 3,4,5 aur 6,8,?

- Step: Ratio = $3/6 = 1/2$ (har side).
- Step: 5 ke corresponding: $5 \times 2 = 10$.

Answer:10

Example 10 (Area): Areas ratio 9:16 ho to sides ratio?

- Step: Sides ratio = $\sqrt{\text{areas ratio}}$.
- Step: = $\sqrt{9} : \sqrt{16} = 3 : 4$.

Answer:3 : 4

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

- Q1.** Triangle ke teeno angles ka sum kitna hota hai?
- Q2.** Similar triangles ki 2 conditions kya hain?
- Q3.** Congruent aur Similar me kya antar hai?
- Q4.** Similarity ke 3 criteria ke naam batao.
- Q5.** BPT (Thales) theorem kya kehta hai?
- Q6.** Pythagoras theorem likho.
- Q7.** Right triangle: base=6, perpendicular=8, hypotenuse?
- Q8.** Kya 5, 12, 13 ek right triangle hai?
- Q9.** Similar triangles ki sides ratio 3:4 -> areas ratio?
- Q10.** AA criterion me kitne angles equal chahiye?
- Q11.** $DE \parallel BC$, $AD/DB = AE/EC$ - ye kaunsa theorem hai?
- Q12.** Hypotenuse=10, base=6, perpendicular?
- Q13.** Do similar triangles ki corresponding sides 4 aur 8 -> ratio?
- Q14.** Areas ratio 9:16 ho to sides ratio kya hai?
- Q15.** Pythagoras theorem kis type ke triangle me lagta hai?
- Q16.** SSS similarity criterion kya kehta hai?
- Q17.** SAS similarity me kya-kya chahiye?
- Q18.** Triangle me sabse lambi side kis angle ke saamne hoti hai?
- Q19.** Similar triangles ka symbol kya hota hai?
- Q20.** Kya 9, 12, 15 right triangle hai?

Step-by-Step Solutions

- Q1.** Teeno angles ka sum?
 - Step: Angle sum property.**Answer:180 degree**
- Q2.** Similar triangles ki 2 conditions?
 - Step: Angles + sides dekho.**Answer:Angles equal + sides proportional**
- Q3.** Congruent vs Similar?
 - Step: Size ka farak dekho.**Answer:Congruent = same size; Similar = size alag**
- Q4.** Similarity ke 3 criteria?

- Step: Standard criteria.

Answer:AA, SSS, SAS

Q5. BPT kya kehta hai?

- Step: Parallel line baaki sides ko same ratio me baantti hai.

Answer: $DE \parallel BC \rightarrow AD/DB = AE/EC$

Q6. Pythagoras theorem?

- Step: Right triangle me.

Answer: $\text{hyp}^2 = \text{base}^2 + \text{perp}^2$

Q7. base=6, perp=8, hyp?

- Step: $\text{hyp}^2 = 36 + 64 = 100$.
- Step: $\text{hyp} = \sqrt{100}$.

Answer:10

Q8. Kya 5,12,13 right triangle?

- Step: $5^2 + 12^2 = 25 + 144 = 169 = 13^2$.

Answer:Haan

Q9. Sides ratio 3:4 $\rightarrow$ areas ratio?

- Step: $(\text{sides ratio})^2 = 3^2 : 4^2$.

Answer:9 : 16

Q10. AA me kitne angles equal?

- Step: Do angles kaafi hain.

Answer:2 angles

Q11. $DE \parallel BC$, $AD/DB = AE/EC$ kaunsa theorem?

- Step: Basic Proportionality.

Answer:BPT (Thales)

Q12. hyp=10, base=6, perp?

- Step: $100 = 36 + p^2 \rightarrow p^2 = 64$.

Answer:8

Q13. Sides 4 aur 8 $\rightarrow$ ratio?

- Step: 4/8 simplify.

Answer:1 : 2

Q14. Areas ratio 9:16 $\rightarrow$ sides ratio?

- Step: $\sqrt{9} : \sqrt{16}$.

Answer:3 : 4

Q15. Pythagoras kis triangle me?

- Step: Right angle wala.

Answer:Right-angled triangle

Q16. SSS similarity?

- Step: Teeno sides ka ratio same.

Answer:Teeno corresponding sides proportional

Q17. SAS similarity me?

- Step: 2 sides + beech ka angle.

Answer:2 sides proportional + included angle equal

Q18. Sabse lambi side kis angle ke saamne?

- Step: Bade angle ke saamne badi side.

Answer:Sabse bade angle ke saamne

Q19. Similar triangles ka symbol?

- Step: Tilde symbol.

Answer:~

Q20. Kya 9,12,15 right triangle?

- Step: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

Answer:Haan